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Intensive Care Outcome Prediction With Machine Learning

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Alla Dott.ssa Maria Regina Morales,
che con pazienza e umanità mi ha accompagnata nel mio risveglio emotivo.
Grazie per avermi aiutata a sciogliere i nodi del passato, a trasformarli
in consapevolezza, e a vivere con gioia i frutti di cio' che ho costruito.

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Acronyms

AE AutoEncoder

AUC Area Under Curve

CSV Comma Separated Values

DFT Discrete Fourier Transform

FFT Fast Fourier Transforms

ICU Intensive Care Unit

LOS Length Of Stay

ML Machine Learning

MV Mechanical Ventilation

NN Neural Networks

PCA Principal Component Analysis

PS Pressure Support

RSBI Rapid Shallow Breathing Index

SBT Spontaneous Breathing Trial

VAE Variational AutoEncoder

WOB Work Of Breath

Abstract

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Chapter 1

Introduction

Mechanical Ventilation (MV) is a life-saving technique used for patients experiencing acute respiratory failure within the Intensive Care Unit (ICU). [3] A large proportion of these patients require MV for more than 24 hours. [6] While MV is life-saving, it is also invasive and carries numerous risks, including baro-volotraumatic lesions, Ventilator-Associated Pneumonia (VAP), airway trauma, and complications related to respiratory muscle fatigue, such as diaphragmatic injury. [6] Given these significant adverse events, MV weaning and endotracheal extubation represent a major challenge during the ICU stay, and demand the attention and expertise of critical care specialist. [6]

The central dilemma in weaning management consists of balancing the risks associated with prematurely extubation (which leads to complications and increased risk of mortality) and the dangers of prolonging MV unnecessarily. [3] [6] Determining the optimal moment for extubation involves extensive expertise, as well as structured clinical protocols. [4] [6] These protocols typically involve daily screening for objective clinical improvement criteria, which establish the pre-test probability of weaning. [6]

If these readiness criteria are met, the next crucial step is the Spontaneous Breathing Trial (SBT), which is defined as a trial of unassisted breathing used to assess whether the patient can sustain respiration on their own. [13] We specifically will focus on Pressure Support (PS) SBTs, which last 30 minutes [9] and during which the patient receives minimal respiratory support. This trial is conducted for a prescribed amount of time, commonly 30 to 120 minutes. [9] The outcome of this SBT is critical, as weaning success is formally defined as extubation followed by the absence of ventilatory support within 48 hours. [13] On the other hand, weaning failure is defined as the inability to pass an SBT or the need for reintubation shortly thereafter.

Predicting and determining weaning failure is so difficult because of its complex, multifactorial nature. In fact, the causes for failed mechanical ventilation (MV) are often related not only to the patient's pulmonary status. Some retrospective studies have found that the most common reason for SBT failure is cardiovascular dysfunction. [3] [6]

Conventional clinical parameters have been proven to have insufficient sensitivity and specificity to accurately predict extubation outcome. Specifically, no single appropriate predictor can be relied upon to determine success, nor can any set of parameters than focuses narrowly on respiration. [6]

This diagnostic uncertainty is compounded by the sheer quantity and high-dimensionality of the data routinely generated in the Intensive Care Unit (ICU), which consists of numerous continuous physiological data streams. [3] These signals are of different nature, and span from vital signs (including pulmonary, cardiovascular, and neurological metrics) and ventilator settings, to demographic information and prognosis. [10] [12] [13]

The magnitude, multitude and complexity of ICU data can cause human caregivers to miss crucial information, that is necessary for precise day-to-day or immediate outcome predictions. [4] To address this limitation of human data processing, Machine Learning (ML) algorithms are employed. ML offers the intrinsic capacity to effortlessly analyze this high-dimensional data. [6] ML algorithms are also able to detect the complex (non-linear) underlying patterns that are distinctive of ICU data and that are tricky to decipher even for trained medical personnel. [3] [4] The resulting ML model is intended to be a useful clinical decision support tool (not a complete replacement for clinical judgment). The guidance provided by this tool in the mechanical ventilation weaning process should improve patient outcomes and facilitate effective use of hospital resources.[3] [13]

Our work addresses this challenge by modeling SBT prediction as a binary classification problem (Success/Failure) aimed at determining the outcome of SBTs. This strategy relies on integrating diverse physiological signals, specifically on 10 selected medical signals including pulmonary and cardiac signals. This multimodal data integration is critical and supported by prior successful ML attempts. [6]

We focused on the role played by the feature engineering techniques used for ML, which provide meaningful representations from these data. The first technique consists of extracting simple, handcrafted features, consisting of the first three central statistical momenta. Since these features were chosen arbitrarily, they could cause information loss and subsequent poor classification performance. For this reason, the other feature engineering strategies consisted of unsupervised dimensionality reduction techniques: Principal Component Analysis (PCA) and a Convolutional Neural Network (CNN) autoencoder architecture. The comparison between these 3 strategies is made possible by utilizing the same Feed-Forward Neural Network (FFNN) architecture for all 3 kinds of features.

The efficacy of these methods is quantified in terms of Area Under the Curve (AUC). ... ADD MORE SPOILERS WHEN DONE WITH EVERYTHING - ABSTRACT-STYLE

This work is structured as follows:

- Chapter 2 dives deeper into the Related Work and medical context;
- Chapter 3 carries out data-preprocessing and SBT extraction, analyzing the results;
- Chapter 4 describes the ML architecture and feature engeneering;
- Chapter 5 illustrates the results of these methods and compares them to one another;
- Chapter 6 comments and draws conclusions on the results obtained in Chapter 5.

Chapter 2

Related Work

Mechanical ventilation (MV) is a critical intervention frequently used for patients experiencing acute respiratory failure within the Intensive Care Unit (ICU).
The clinical objective is to wean the patients at the optimal moment [6], balancing between the risks associated with premature extubation (e.g., significant risk of increased morbidity and mortality [6], pulmunary and airway risks, cardiovascular and hemodynamic risks, neurological and mental status risks [8]) and the risks of unnecessarily prolonged MV (e.g., infections, baro-volotraumatic lesions [6], diaphragmatic injury [4]).

Choosing the appropriate time for secure extubation requires structured clinical protocols, which involve the daily screening for objective clinical improvement criteria.
Once these criteria are met, the next step is performing a Spontaneous Breathing Trial (SBT, i.e. a period of unassisted breathing) to assess the patient's ability to sustain ventilation independently.[6] [13] Weaning success is formally defined as successful extubation followed by the absence of ventilatory support within a specified timeframe, commonly 48 or 72 hours. [1][6] Therefore, just passing a SBT does not guarantee weaning success.

Weaning difficulty is typically categorized into three internationally recognised groups based on the number of SBTs executed before successful extubations:[1] [6] [13]

- Simple Weaning: patients successfully complete the SBT and are immediately extubated. This group represents approximately 60% of weaned patients and is associated with a favorable prognosis (5% ICU mortality). [6] [13]
- Difficult Weaning: patients require one to three SBTs before successful extubation, all within a period of less than seven days after the first attempt [4] [6]
- Prolonged Weaning: patients require more than three SBTs or remain under MV for more than seven days after the initial attempt, before final extubation. [6] The underlying causes often involve pronounced contraction weakness of the diaphragm combined with deconditioning of the peripheral respiratory musculature.[4]

The clinical outcomes for patients in the difficult and prolonged weaning groups are significantly worse, with reported ICU mortality rates equal to or higher than 25%.[6]
Overall, failure of endotracheal extubation still occurs, in $15-20\%$ of cases, despite careful clinical protocols. [9] [3]

To improve the success rate of MV weaning, several systematic reviews and prospective studies have been carried out to identify the optimal moment of extubation. [10] Histor-

ically, research concentrated on comparing different weaning techniques (such as T-piece trials versus Pressure Support Ventilation (PSV)) and establishing the utility of traditional weaning indicators. However, traditional clinical parameters (e.g., Rapid-Shallow Breathing Index (RSBI)) have shown insufficient sensitivity and specificity to accurately predict extubation outcome. [2] Specifically, no single predictor can accurately determine weaning success, as failure is complex and multifactorial, as the pulmunary status can be affected by cardiovascular or neurological issues. [6]

The difficulty in relying solely on traditional methods is compounded by the sheer volume and complexity of high-dimensional data generated in the ICU, which consists of numerous continuous physiological signals. Human caregivers may inadvertently miss crucial information or hidden patterns necessary for precise outcome predictions due to this magnitude and the multitude of influencing factors. [3] [4]

To address these limitations, Machine Learning (ML) algorithms are increasingly being utilized. [13] ML offers the intrinsic ability to analyze high-dimensional data effortlessly and detect the complex underlying correlations that are characteristic of ICU data. Research efforts have thus focused on developing dynamic and comprehensive predictive models using ML/AI to improve clinical decision-making during the weaning process. [3] [6] [13] In published literature, ML has been applied to various patient populations undergoing weaning and SBTs

One of the first applications of ML to ICU data was limited to neonatal ICU patients. In 2017, [8]'s developed ML models to predict extubation readiness in extremely preterm infants (gestational age $\leq$28 weeks). The ML features were obtained from cardio-respiratory signals and clinical variables using automated analysis systems (AUREA), and were fed to supervised ML classifiers such as Support Vector Machines and AdaBoost. The model is validated using cross-validation and ROC analysis and includes risk stratification (success, fail, uncertain). A pilot study showed that combining 17 features using SVM produced promising classifications, indicating the potential to significantly reduce extubation failures in this vulnerable population.

Soon after, in 2018, [5] developed and compared multiple machine learning algorithms for predicting ICU mortality. The authors collected demographic, clinical, severity-score data and comorbidities from 341 unplanned extubation patients admitted between 2008 and 2017. After preprocessing, the data was splitting between training and test sets, of 307 and 34 patients respectively. Out of the four predictive models (artificial neural network, logistic regression, random forest, and support vector machine), the random forest model achieved the best performance (precision = 0.882, AUROC = 0.910) and significantly outperformed conventional severity scores such as APACHE II and TISS. Hyperparameters for each algorithm were tuned via 10-fold cross-validation; class imbalance was handled by weighting the training instances according to outcome ratio.

A few years layer, in 2023, [6] used ML to identify predictors of successful weaning from MV in adult ICU patients. This prospective application integrates hemodynamic, respiratory, and heart-rate variability (HRV) signals from 108 patients across 135 spontaneous breathing trials. The dynamic predictive model is obtained by exploring nonlinear associations among numerous physiological variables without predefined hypotheses using symbolic regression and evolutionary genetic algorithms (ZGPD model, MyDataModels).

The final predictive model uses pre-SBT and intra-SBT data and includes body mass index, occlusion pressure at 0.1 s (P0.1), HRV ratio (LF/HF), and heart rate during SBT, achieving up to 83 % accuracy for predicting 72-hour extubation success.

In 2024, [12] The study focuses on developing a ML model to predict the risk of prolonged mechanical ventilation (PMV, > 14 days of intubation), in patients admitted to an ICU with COVID-19. It uses clinical data available at the time of intubation from a single academic hospital (112 patients) including routinely measured parameters. The dataset was split into two chronologically distinct training and test set. This strategy achieved a respectable performance on the test set (AUROC= 0.804).

That same year, [13] developed a ML tool to identify ICU patients on mechanical ventilation who are at high risk of difficult weaning. To achieve this, the researchers conducted a retrospective cohort study of 487 adult ICU patients (of which 323 underwent "simple weaning" and 164 "difficult weaning"). They collected 36 variables including demographic data, severity scores like APACHE II and SOFA, physiological and ventilator parameters, blood gas and lab results. The dataset was split 70/30 between training and test set. The analysis compared five machine-learning algorithms—logistic regression, support vector machine (SVM), random forest, Light GBM and XGBoost. Model performance was assessed and the random forest model performed best (AUC=0.805, accuracy=0.748).

Also in 2024 [4] developed a day-to-day, data-driven predictive ML model specifically for prolongued weaning patients, due to the high complexity and high mortality associated with it. The retrospective dataset included of over 18,000 patient-days from a specialized weaning unit. They use histogram-based Gradient Boosting models (both classifiers and regressors) to predict the duration of unassisted spontaneous breathing (SBT duration) on the following day, based on 61 clinical variables (including vitals, ventilator settings, labs, medications, and fluid balance). The algorithm was optimized via cross-validation on high-performance computing infrastructure, so it was able to handle large-scale inputs. The best model achieved an ROC-AUC of 0.713.

Our study is a retrospective analysis based on a broader and larger SBT dataset than most than most prior weaning-specific ML studies: it includes all three weaning-difficulty groups from the general ICU population, rather than a narrow clinical context (such as COVID-19 or trauma). Unlike studies focused on long-term or dynamic outcomes, we target the immediate SBT result using short pre-SBT physiological windows, which positions us closer to the clinical decision point. To support practical integration into ICU workflows, we limit our dataset to a small set of key physiological signals that remain approximately stationary during brief pre-trial segments (sometimes as short as 15 minutes) to avoid continuous high-frequency monitoring or computationally intensive processing. Methodologically, we adopt a deliberately minimalist modeling architecture, using a compact feed-forward neural network to isolate the role of feature representation, which we explore across three variants: handcrafted features, PCA-derived components, and AE/VAE latent embeddings. Our training and evaluation procedures follow established best practices: model performance is assessed using AUC-ROC, precision, and accuracy; hyperparameters are optimized via k-fold cross-validation on a validation set; class imbalance is addressed through SMOTE for training and validation; and final results are reported on an unbalanced test set, which also informs the selection of the optimal decision threshold.

Chapter 3

Data Preprocessing

The dataset used was provided by the ICU of the Teaching Hospital of the University of Padova. The dataset consists of anonymous csv files, each one covering up to 12 hours of medical signals of a single patient. Each csv file contains, other than the interesting medical signals, the date and time of the measurement, the bed that the patient was in, and the anonymous numerical identifier assigned to the patient, from from September 29th, 2023 to January 26th, 2025.

In reality, the anonymous numerical identifier assigned to the patient is often missing. This is a direct consequence of a systematic delay in assigning the anonymous patient identifier. Specifically, when a patient is first admitted to the ICU, its data is recorded anonymously, until the admission is officially recorded in the system and the patient is assigned the numerical identifier. Sometimes, this association happens only by name, which was erased in our anonymous version of the dataset, resulting in entire stays that cannot be traced back to a patient identifier. For this reason, from here on, we will refer to patients with missing numerical identifier as **untraceable**, and to patients with a numerical identifier as **traceable**.

In the following sections, we will analyze the initial dataset and implement data reconstruction techniques to compensate for the presence of a substantial amount of files lacking meaningful numerical identifiers, as well as to even out the time-series structure of the data by interpolating over the gaps on a uniform grid based on the sampling frequency.

During all preprocessing stages, we focus on matching individual records in the datasets to longer or multiple patient stays, since patients can return to the ICU for different reasons. Separating data by stay (instead of by patient) guarantees the independence of different ICU stays, which can correspond to different prognosis. This is implemented by grouping csv files into a stay if they share the same patient ID and if the temporal distance between each other is less than a threshold $\delta = 6$ hours. This threshold was selected after consulting with clinical personnel as a realistic value, big enough to cover any gaps due to medical exams, but small enough to avoid compensating for entire missing csv files (which we properly correct in data reconstruction).

Taking care of file overlaps was also addressed at all preprocessing stages. The most significant (longest) overlaps happen during bed transfer inside the ICU, where the patient is partially attached to both ICU monitors. Therefore, two files are generated at the same time, without guarantees on the specifical medical signals being recorded. These repetitions amongst files are fixed by finding the overlaps (among files of the same patient), and keeping the data with the most complete set of relevant signals (see Table (3.1)).

3.1 Qualitative Selection of Medical Signals

The first step in data preprocessing consists of selecting the most important medical signals for identifying SBTs. The selection was carried out based on the expertise of the supporting medical team. The interesting medical signals are shown below, together with some quantities necessary for the data analysis.

Name	Unit/Format	Meaning	Abundance
P0.1	cmH_2O	airway obstruction pressure at 100 ms	48.08%
RSBI	breaths/(min/L)	rapid shallow breathing index	31.72%
Art-S	mmHg	arterial systolic pressure	90.57%
SpO_2	%	peripheral oxygen saturation	95.65%
SpO_2 PI	%	perfusion index	95.75%
HR	beats/min	heart rate	96.27%
Cdyn	ml/cmH_2O	dynamic compiance	52.52%
PIP	cmH_2O	peak inspiratory pressure	54.91%
fspn	respirations/min	spontaneous respiratory frequency	54.91%
VTe	ml	expiratory tidal volume	54.91%
WOB	J/L	work of breathing	52.61%
PatientID	integer	anonymous numerical identifier of patient	61.94%
Bed	integer $\in [1, 12]$	bed occupied by patient	100%
Date	dd/mm/yyyy	date of recording	100%
Hour	hh	hour of recording	100%
VWM	string	ventilator work mode	92.26%

Table 3.1: Name, unit of measurement, meaning and abundance for each interesting medical signal and of the other interesting quantities used in the analysis.

At this point in the preprocessing, this qualitative selection was used only to discard the excess columns[1] and to reduce the amount of data. Later on[2], a more in-depth analysis is carried out to impose the columns that a file needs to have in order to be kept.

As previously mentioned, the files was grouped into stays to gather more insight, and the results are shown in Table (3.2) and in Figure (3.1).

The high number of short, untraceable stays (in one bed, consisting of one file) is consistent with these files lacking patient identifier and not being able to be patched together. The few untraceable stays with more than one file come from a few non-numerical identifiers, which were corrected, but were still unreliable and labeled as untraceable.
On the contrary, traceable files correctly have longer stays, made of numerous files, and can occupy multiple beds.

[1]see Table (6.1) from 6 **Appendix: List of all medical signals available**
[2]See 3.3 **Qualitative selection of medical signals**.

	total	traceable	untraceable
duration [days]	4609.4	3129.6	1479.8
number of stays	3747	1077	2670

Table 3.2: Total number and duration of the stays, calculated on the original dataset.

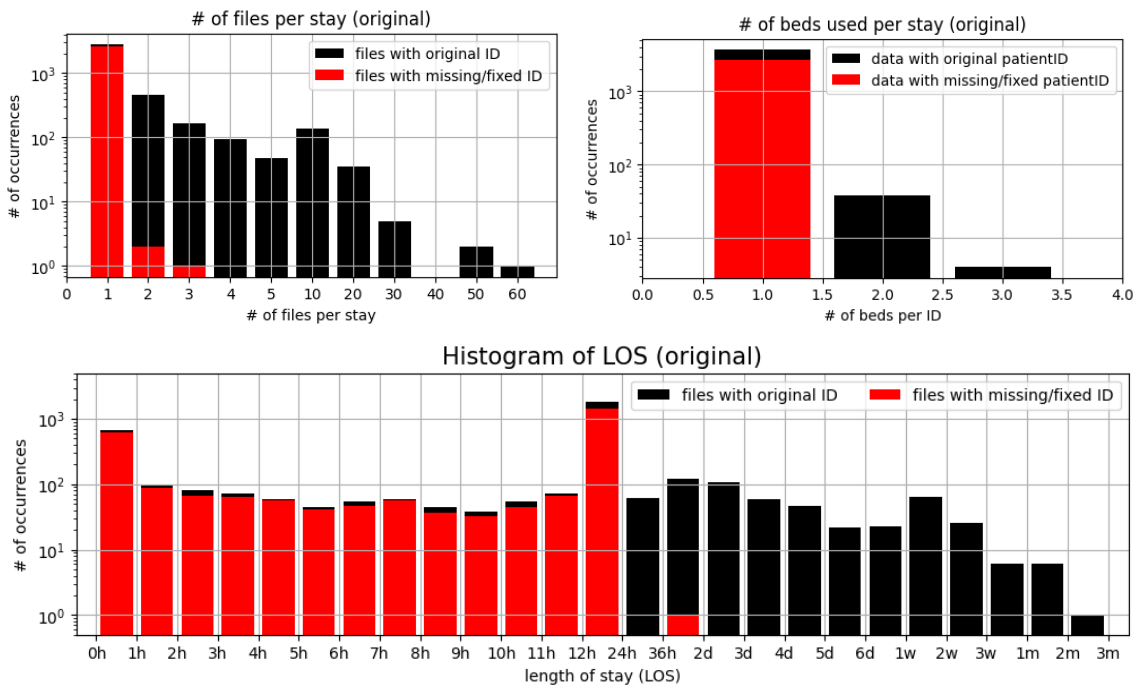

Figure 3.1: Histograms of the number of files, number of beds, and duration (Length of Stay) of the stay of the original dataset.

3.2 Dataset Reconstruction

As previously mentioned, data reconstruction is necessary due to the high amount of untraceable files, which cannot be grouped into stays as they are.
For this reason, in subsection (3.2.1), we will patch together files into stay to minimize the number of untraceable files. Afterwards, in subsection (3.2.3), we will focus on homogenizing the time-series aspect of the data, finding gaps and interpolating over them, using the sampling frequency estimated in subsection (3.2.2).

3.2.1 Merging Files per Patient and Bed

Since these untraceable files are caused by systematic delays in labeling, we performed data reconstruction by merging files if they belong to the same bed and if their temporal distance between their extremities between is smaller than a threshold ε. The threshold $\varepsilon = 30$ min was set as a conservative estimate of the ICU bed turnover time (i.e., the time necessary to get the bed ready for another patient).

The data reconstruction algorithm can be easily visualized in the occupancy charts of the ICU beds over time, in Figure (3.2).

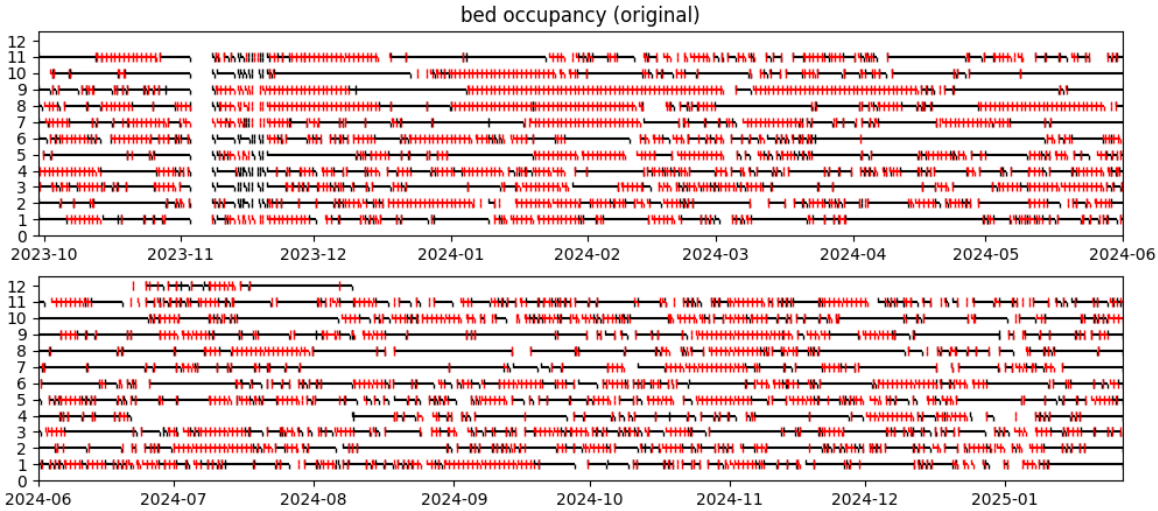

Figure 3.2: Bed occupancy chart before reconstruction, highlighting the difference between traceable (in black) and untraceable (in red) files.

The original occupancy chart (in Figure (3.2) shows a substantial amount of untraceable files, which are characterized by very short duration and are located close to one other.

The data reconstruction is carried out in two steps, in order to optimize the amount of traceable data. First, traceable data is patched with untraceable data, with the goal of tracing back as many untraceable files as possible to meaningful anonymous identifiers. Specifically, at this step we patch traceable files with untraceable files if they share the same bed and if their extremities differ by less than the threshold ε.

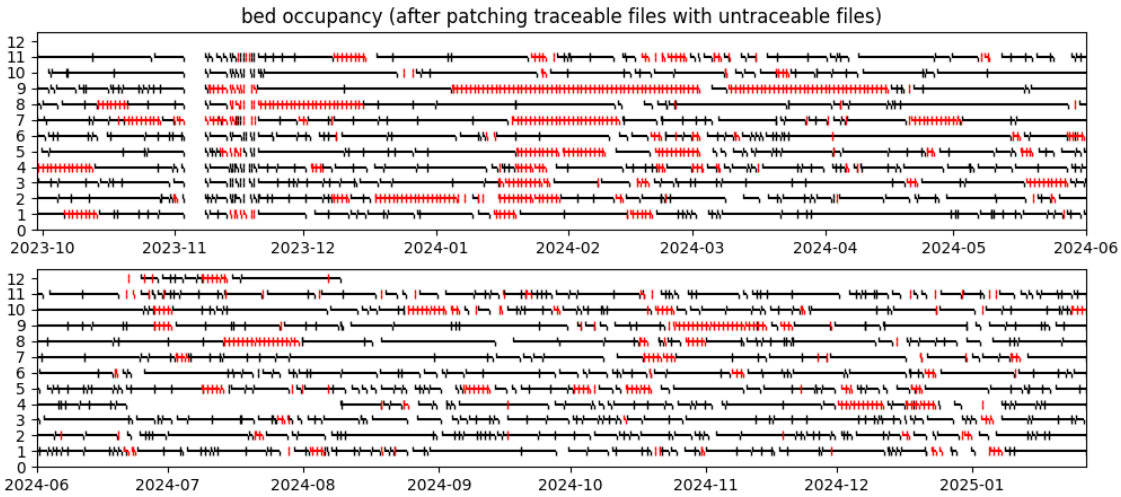

Figure 3.3: Bed occupancy chart after the first step of patching (traceable files with untraceable files), highlighting the difference between traceable (in black) and untraceable (in red) files.

As shown in Figure (3.4) and Table (3.3), this first patching step results in a significant decrease in number of untraceable files and a significant decrease of the number of ICU

stays overall. This also results in even longer durations and higher the number of files and beds of traceable stays.

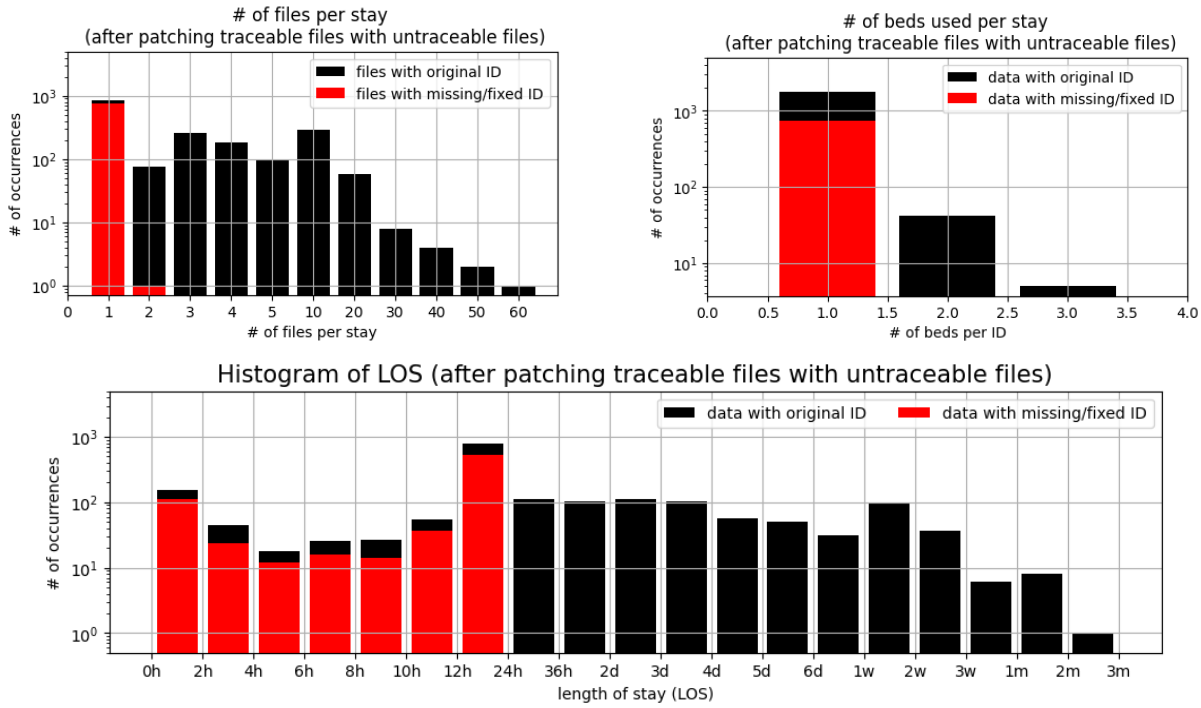

Figure 3.4: Histograms of the number of files, number of beds, and duration (Length of Stay) of the stay after the first patching step.

The second and last patching step consisted of patching amongst the remaining untraceable data: this step is carried out to ensure that the remaining untraceable files have longer durations, comparable to the ones of traceable stays. Specifically, at this step we patch together untraceable files that share the same bed and if their extremities differ by less than the threshold ε.

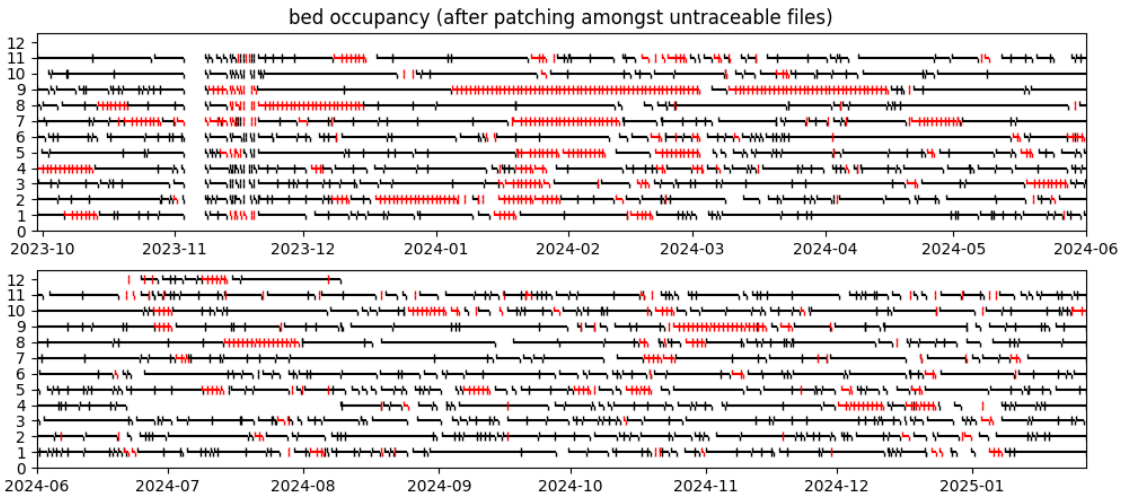

Figure 3.5: Bed occupancy chart after all patching (including patching amongst untraceable files) highlighting the difference between traceable (in black) and untraceable (in red) files.

The effects on the dataset of this second patching step are more subdued. It's hard to notice any difference between the before (in Figures (3.3) and (3.4)) and the after (in Figures (3.5) and (3.6)). The only noticeable difference is a small decrease in the number of untraceable files, as shown in Table (3.3) which is compatible with the desired merging of untraceable stays. We also notice that the number of traceable stays is correctly constant between these two steps.

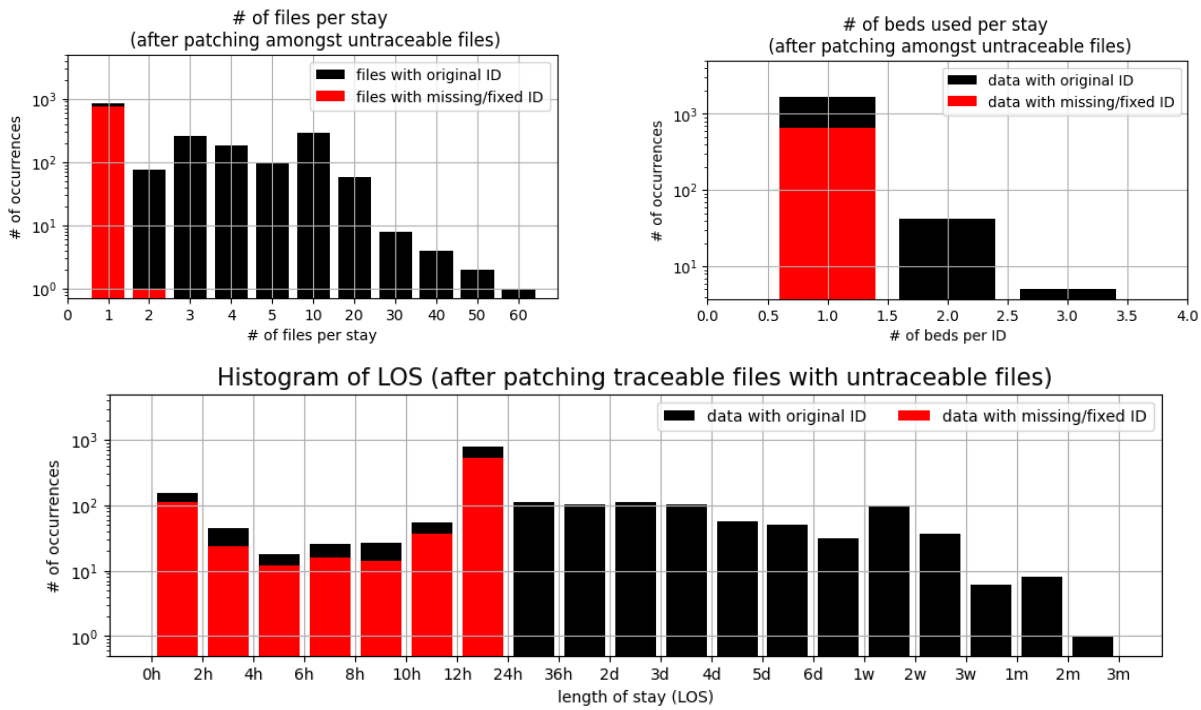

Figure 3.6: Histograms of the number of files, number of beds, and duration (Length of Stay) of the stay after all patching steps.

duration [days]	total	traceable	untraceable
original	4609.4	3129.6	1479.8
after first step of patching	4591.5	4057.7	533.7
after second step of patching	4591.5	4057.7	533.7
after patching + min duration	4590.7	4057.5	533.2
number of stays	total	traceable	untraceable
original	3747	1077	2670
after first step of patching	1822	1076	746
after second step of patching	1726	1076	650
after patching + min duration	1680	1044	560

Table 3.3: Evolution of the total number and total duration of the stays, before and after the two steps of patching, and after selecting for the minimum duration.

The resulting dataset of stays is then screened to ensure a minimum duration that is compatible with:

- the extraction of a pressure support SBT, which lasts 30 minutes [9]
- our ML approach, which examines the pre-trial data to predict the SBT outcome and needs a fixed pre-trial duration, which we set arbitrarily to 30 minutes

The total minimum duration is 1 hour. As we can see below in Figure (3.7) and in Table (3.3), the number of stays with minimum 1 hour duration is still high, especially after all these patching steps.

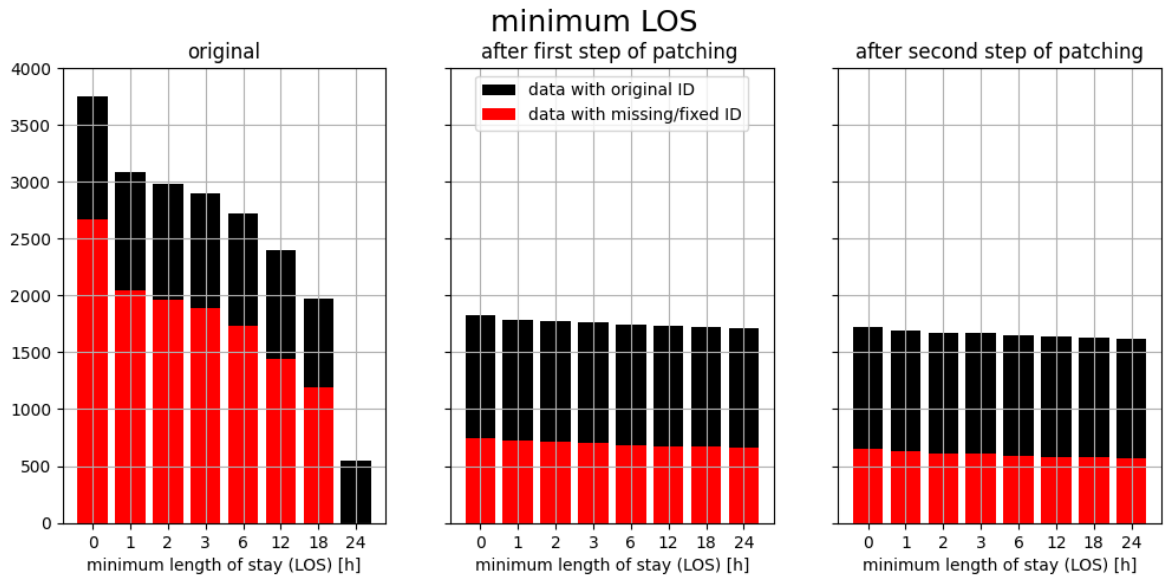

Figure 3.7: Evolution of the duration of time [days], before and after the two steps of patching.

The resulting dataset is a set of all patient stays in the ICU that are longer than 1 hour. The information about each stay, which was obtained by merging files, is likely to show gaps or overlaps. Therefore, the next preprocessing step will consist of regularizing the timeseries that constitute our stays.

3.2.2 Sampling Frequency Estimation

Estimating the sampling frequency of the data is necessary to guarantee an uniform structure of the data. To achieve this goal, we first compute the sampling period τ_i of each file i as the mode of the time differences.

$$\tau_i = \arg\max_t \; p(\Delta_t) \tag{3.1}$$

The sampling period of the whole dataset is computed as the mode over the files:

$$\tau_s = \arg\max_i \; p(\tau_i) = 30\,\mathrm{s} \tag{3.2}$$

The sampling frequency f_s is then computed as the reciprocal of the sampling period τ_s.

$$f_s = \frac{1}{\tau_s} = \frac{1}{30\,s} = 0.0\overline{3}\ \text{Hz} \tag{3.3}$$

As we can see in Figure (3.8), the number of files (8) with sampling period τ_i different to the overall sampling period τ_s is very small (NN), so they were discarded.

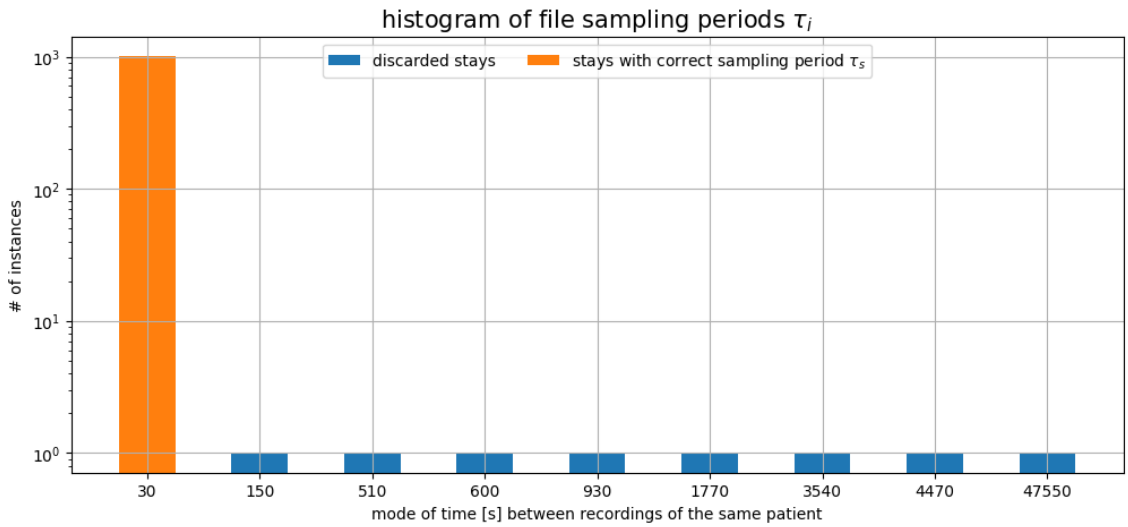

Figure 3.8: Histogram of the sampling periods computed over the stays. The sampling period of the whole dataset is highlighted in orange, and the remaining stays, with different sampling period (which are discarded), are highlighted in blue.

3.2.3 Interpolating Over Gaps

Knowing the sampling frequency of the data allows to recognize where Estimating the sampling frequency of the data is needed for:

- recognizing missing rows in the data
- adding the missing rows to ensure uniform time increases in our timeseries data
- interpolating over the missing data (both missing values and rows), either:
  - linearly with time, for internal gaps and numerical features
  - using backwards or forward filling, for cathegorical features or when the missing data is located at the very beginning or at the very end of the stay

At the end of the preprocessing, out dataset is comprised of fully-reconstructed timeseries, representing patient stays and of minimum 1 hour duration. These timeseries still contain a variable number of medical signals; this limitation will be addressed quantitatively in the next section.

3.3 Quantitative Selection of Medical Signals

The following analysis is aimed at setting which medical columns to keep in our timeseries. Specifically, we would like to minimize the number of medical signals used, since the ML features will be extracted from them, and it's less computationally expensive to train a model with less features. To compromise between this need and the medical team expertise, we would like to reduce the initial list of important medical signals in Table (3.1). Two strategies were used as exploratory tools to quantitatively examine the set of physiological variables considered up to that point.

Firstly, we analyze the abundance fo the medical signals: we observe that, for both traceable and untraceble stays, around half of the stays are missing from around a third of the files. Specifically, the least abundant signal is RSBI.

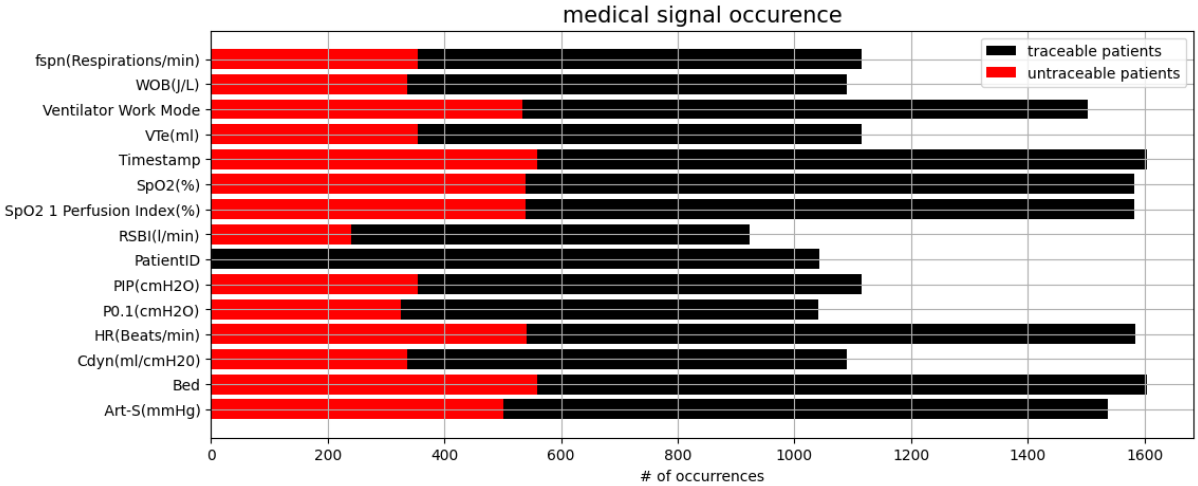

Figure 3.9: Histogram of the medical signals, highlighting the traceable stays (in black) and the untraceable stays (in red).

We also applied Principal Component Analysis (PCA) to the numerical signals in our dataset to assess which variables contributed most strongly to the dominant variance directions. This helped identify variables that were highly correlated, variables that had little influence on overall variation, and potential redundancies in the expert-selected signal set.

Since PCA is variance-sensitive (i.e, features with larger ranges will dominate the analysis and skew the results), PCA was performed over standardized inputs $Z \in \mathbb{R}^{n \times p}$ obtained from the original input $X \in \mathbb{R}^{n \times p}$ as:

$$z = \frac{x - \mu_x}{\sigma_x} \quad \forall x \in X \quad \text{where} \quad \mu_x = \frac{1}{|X|} \sum_{x \in X} x \quad , \quad \sigma_x = \sqrt{\frac{1}{|X|} \sum_{x \in X} (x - \mu_x)^2} \tag{3.4}$$

where n is the number of data points and p is the number of features. The standardized inputs Z are used to compute the covariance matrix $\Sigma \in \mathbb{R}^{p \times p}$. The amount of variance explained by each direction v_i is given by the corresponding eigenvalue λ_i of Σ:

$$\Sigma = \frac{1}{n-1} Z^T \cdot Z \tag{3.5}$$

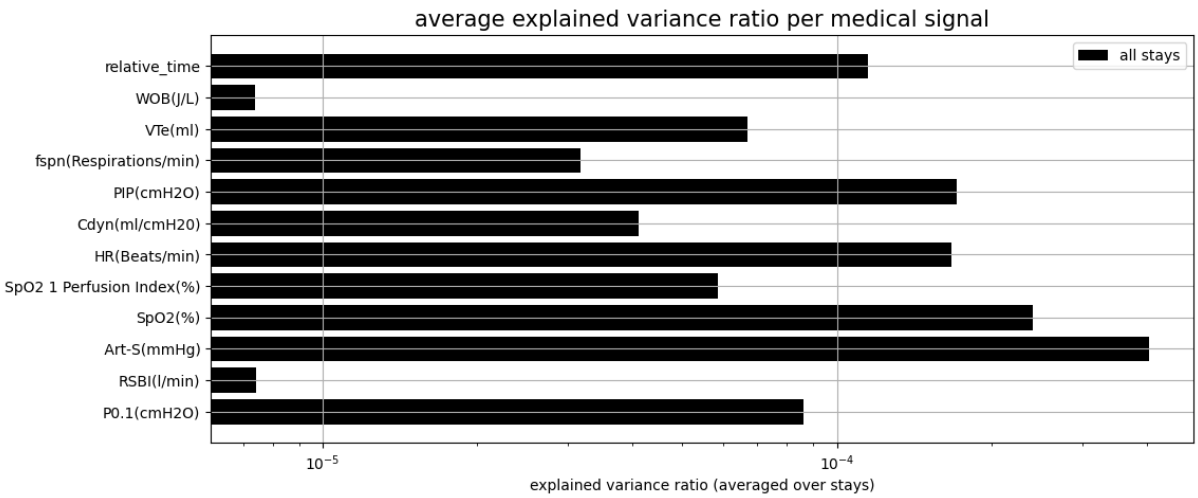

Figure 3.10: Average explained variance ratio of each medical signal (all stays).

In Figure (3.10), we observe that the least important columns are the RSBI and the WOB. The fact that the RSBI has one of the smallest explained variance ratios is justified by the linear dependence that exists between RSBI, fspn and VTe:

$$RSBI(l/min) = \frac{fspn(Respirations/min)}{VTe(ml)} \cdot 1000(ml/l) \tag{3.6}$$

Given that the RSBI is the least present and least important medical signals, we looked into getting rid of it. On advice of the medical team, we instead discarded the VTe, which still solves our redundancy issue.
At the same time, we tried to avoid loosing information in the cases where the VTe is present but the RSBI is not. So, before getting rid of the VTe, we used it to reconstruct the RSBI values, based on the following formula:

$$RSBI(l/min) = \frac{fspn(Respirations/min)}{VTe(ml)} \cdot 1000(ml/l) \tag{3.7}$$

This step successfully recovered data both partially (line by line, in 916 stays) and fully (for full stays, in 191 stays). We also tried fixing the $fspn$ through the inverse formula:

$$fspn(Respirations/min) = \frac{RSBI(l/min) \cdot VTe(ml)}{1000(ml/l)} \tag{3.8}$$

but no improvement was obtain, indicating that all files with VTe and RSBI already contained fspn.

After recovering as many RSBI signals and discarding the VTe signals, we were able to keep 1005 stays (out of the original 1604) which contain all chosen medical signals, which are listed in Table (3.4).

Name	Unit/Format	Meaning	Abundance
P0.1	cmH_2O	airway obstruction pressure at 100 ms	48.08%
RSBI	breaths/(min/L)	rapid shallow breathing index	31.72%
Art-S	mmHg	arterial systolic pressure	90.57%
SpO_2	%	peripheral oxygen saturation	95.65%
SpO_2 PI	%	perfusion index	95.75%
HR	beats/min	heart rate	96.27%
Cdyn	ml/cmH_2O	dynamic compiance	52.52%
PIP	cmH_2O	peak inspiratory pressure	54.91%
fspn	respirations/min	spontaneous respiratory frequency	54.91%
WOB	J/L	work of breathing	52.61%
PatientID	integer	anonymous numerical identifier of patient	61.94%
Bed	integer $\in [1, 12]$	bed occupied by patient	100%
Date	dd/mm/yyyy	date of recording	100%
Hour	hh	hour of recording	100%
VWM	string	ventilator work mode	92.26%

Table 3.4: Name, unit of measurement, meaning and abundance for each chosen medical signal and of the other interesting quantities used in the analysis.

3.4 Filtering & Smoothing

Physiological measurements recorded in the ICU are often affected by noise and occasional extreme values. These imperfections can negatively impact downstream preprocessing and model performance, so several filtering strategies are applied to improve signal quality. This section describes the three filtering methods considered in this study: FFT-based filtering, moving average smoothing, and percentile clipping.

3.4.1 Fast Fourier Transform

We considered using Fourier Transform based filtering, which can be used to transform a digital signals into the frequency domain. Fourier filtering techniques are especially useful for removing harmonic noise, since harmonic patterns are typically found in localized discrete parts of the Fourier transform.

We specifically used the Fast Fourier Transform (FFT, in Python's `scipy.fft` implementation) which reduces computational cost from $O(N^2)$ to $O(N \log(N))$. We resorted to Discrete Fourier Transforms (DFTs) because, although physiological signals are continuous in nature, all signals are digitally recorded at given sampling frequency f_s and therefore represented as discrete time series.

Given a sample $x = [x_0, \ldots, x_{N-1}]$, the direct Discrete Fourier Transform is defined as:

$$\hat{X}_k = \sum_{n=0}^{N-1} x_n \, e^{-i2\pi kn/N} \qquad k = 0, \ldots, N-1. \tag{3.9}$$

and the inverse descrete Fourier transform is defined as:

$$\widetilde{x}_n = \sum_{n=0}^{N-1} \widetilde{X}_k \, e^{+i2\pi kn/N} \qquad k = 0, \ldots, N-1. \tag{3.10}$$

where the coefficients $\hat{x}_k$ represent the contribution of the frequency component $\frac{k}{N}$ (in normalized units) to the signal.

For filtering purposes, we are interested specifically in the frequency spectrum, which is the set of magnitudes $\{|X_k|\}_k$ corresponding to frequency $f_k = \frac{k}{n} f_s$. In the spectra, low frequencies represent slow physiological trends, whereas high frequencies typically are typically associated to noise or sensor jitter. Based on these observations, we can modify the amplitude of selected frequency components by applying a frequency mask H_k:

$$\widetilde{X}_k = H_k \cdot X_k \tag{3.11}$$

before reconstructing the signal through inverse DFT. Depending on the mask, you can filter out high-frequency noise (low-pass filtering), slow baseline drift (high-pass filtering), or narrowband electrical interference (notch filtering).

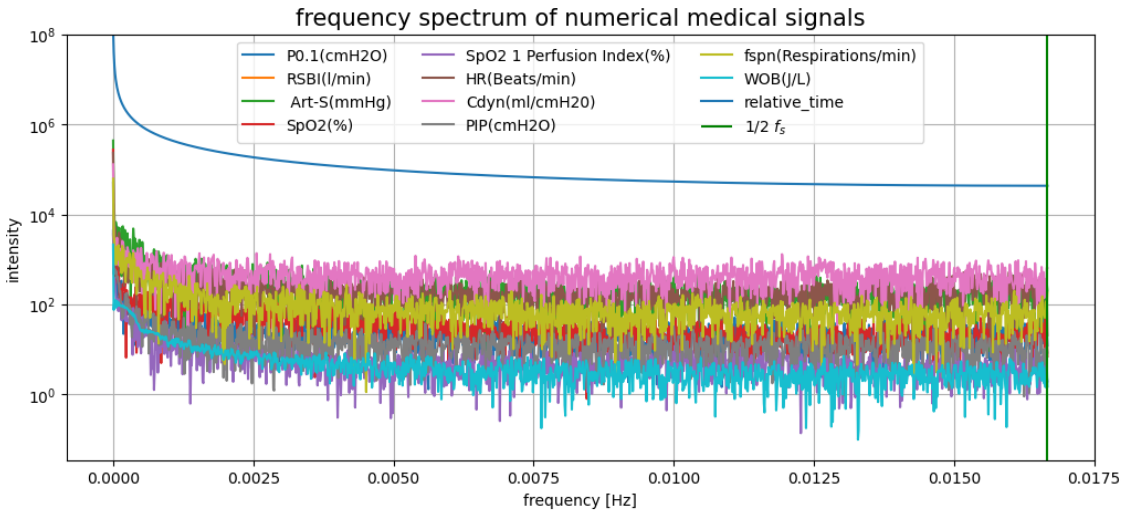

Figure 3.11: Frequency spectra of each medical quantity analyzed for one stay, highlighting the end of the spectrum.

The spectrum of all numerical column was computed and shown in figure (3.11). This spectrum covers a frequency range that depends on the sampling frequency f_s:

$$(0, \frac{1}{2} f_s) = (0, 0.01\overline{6}) \tag{3.12}$$

When dealing with medical data, it's common for data to be noisier due to respiratory and cardiac interference. The normal frequency ranges for these oscillations are:

- resting respiratory range: 12-120 bpm $\sim$ 0.2-2.0 Hz [7]
- sleeping heart rate range: 30-220 bpm $\sim$ 0.5-3.$\overline{6}$ Hz [11]

Therefore, we should filter out the union of these two ranges, (0.2, 3.$\overline{6}$) Hz. Yet, this range is already outside of the frequency range covered by our spectrum.
As a result, we were not able to apply Fourier Transform based filtering.

3.4.2 Moving Average Filter

A simple moving average filter is a simple, non-parametric smoothing approach. For 1D time series, it consists of a convolution with a 1D kernel F of size k. As the name suggests, it corresponds to the unweighted average of a patch of k consecutive data points, taken symmetrically, of the input time series $\{x_i\}_{i=0}^{N-1}$.

A moving average is commonly used with time series data to smooth out short-term fluctuations and highlight longer-term trends or cycles. For this reason, a moving average filter is viewed, in signal processing, as a low-pass finite impulse response filter.
For the reasons listed above, we used a simple moving average filter, with filter size 3.

3.4.3 Percentile-based Clipping

Percentile clipping is a a data-cleaning technique used to reduce the influence of outliers in a dataset. Percentile clipping is especially useful when the dataset contains outliers due to measurement errors or, and because of its non-parametric approach.

This technique is based on two thresholds, typically the lower p-th percentile and the upper $(100 - p)$-th percentile: all observations below the lower threshold and above the upper threshold are discarded, while values that fall between the two percentiles remain unchanged.

In standard implementations, the discarded values are set equal to the lower or upper thresholds. We chose to interpolate over the discarded values, which yields similar results, and used $p = 1$. This results in a simple and robust way of controlling the influence of extreme values.

3.5 SBT Extraction

The SBTs can theoretically be detected based on the Ventilator Work Mode (VWM), which is set to Pressure Support Continuous Positive Airway Pressure (PS-CPAP) during SBTs. The outcome of SBTs can also theoretically be determined based the VWM setting after the end of the SBT, based on Table (3.5).

VWM after SBT end	**outcome**
Standby	Standby Success
Unknown mode	Standby Success
NIV NAVA	Non Invasive Success
NIV PS	Non Invasive Success
NIV PS/CPAP	Non Invasive Success
NAVA	Non Invasive Success
others	Failure

Table 3.5: Definition of the possible SBT outcomes based on the VWM value at the end of the SBT.

In real life, doctors often leave the patient PS-CPAP mode even after establishing the end and outcome of the SBT. Therefore, we cannot reliably extract SBTs based on only on the VWM. An alternative strategy consists of reconstructing the start of the SBTs from the VWM, but establishing the end and outcome of the SBT based on the RSBI values being over a threshold of 105 breaths/min/l. [14]

We built upon the previous criterion by looking for only one SBT for each period of time in which the VWM is in PS-CPAP mode, in order to avoid redundancies. We also imposed a maximum SBT duration of 30 minutes, given that these SBTs are of the pressure-support kind. [9] Any SBTs that do not fail before the 30-minute threshold are labeled as successes, which also establishes their end.
Additionally, instead of applying the RSBI criterion instantaneously (which is sensitive to noise and can lead to an anomalous number of failures), we implement it over a window of 10 values. The SBT fails only when n out of the previous 10 values exceed the RSBI threshold. This reduces the number of failures and more closely reflects clinical practice, where a doctor generally waits for RSBI values to stabilize above the threshold before deeming the SBT a failure and ending it.

For ML purposes, we are interested in saving also the pre-SBT data: for this reason, we discarded SBTs with pre-SBT duration smaller than 15 minutes.

3.5.1 Tuning the failure criteria

As mentioned previously, the failure criteria is based on a fraction $n/10$ of the previous RSBI values being over 105 breaths/min/l. The value of n was chosen to match (as much as possible) the actual success rate of PS SBT, which is equal to 82.3 % [9].

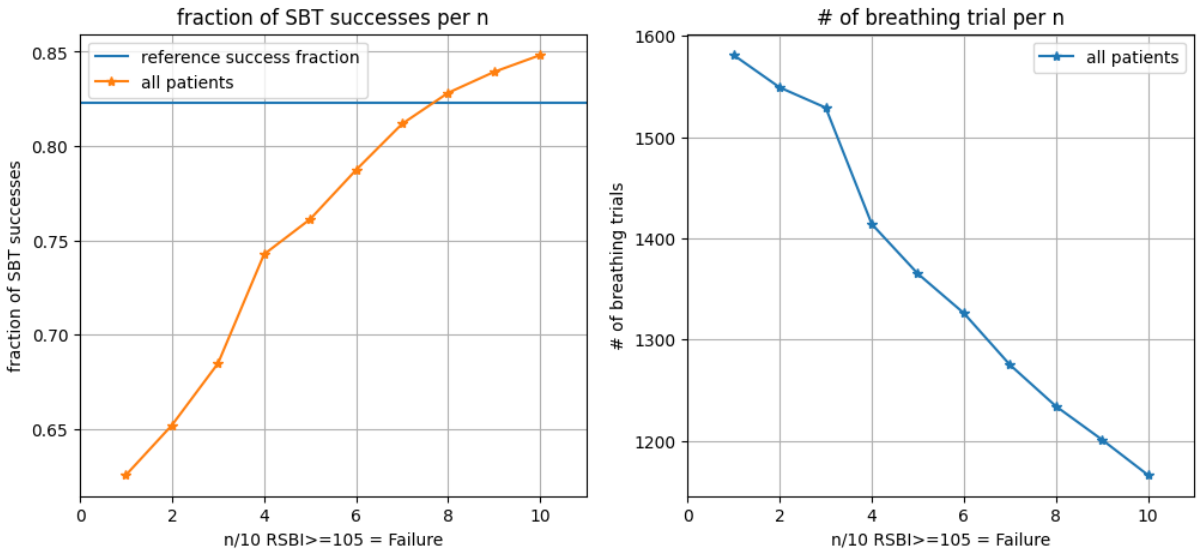

Figure 3.12: Evolution with n of the fraction of SBTs successes and of the number of SBTs found.

3.5.2 Characterization of the SBT dataset

The resulting SBT dataset consists of 1234 SBTs (1022 successes and 212 failures, and 1067 traceable and 167 untraceable).

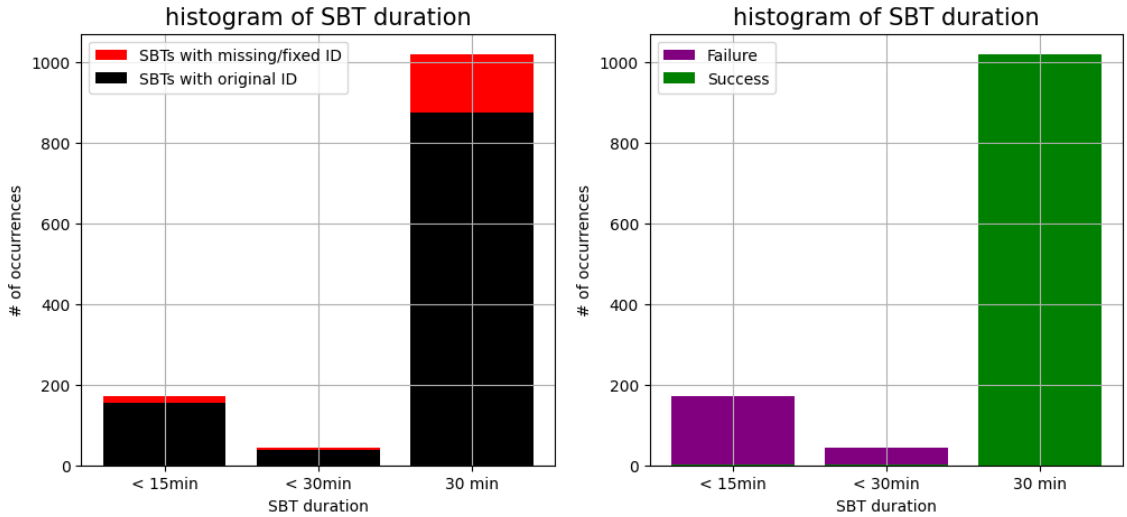

Figure 3.13: Evolution with n of the fraction of SBTs successes and of the number of SBTs.-

As we can see in Figure (3.13) and in Figure (3.14), almost all SBTs that end before 30 minutes are labeled as failures, and all that end after are labeled as successes.

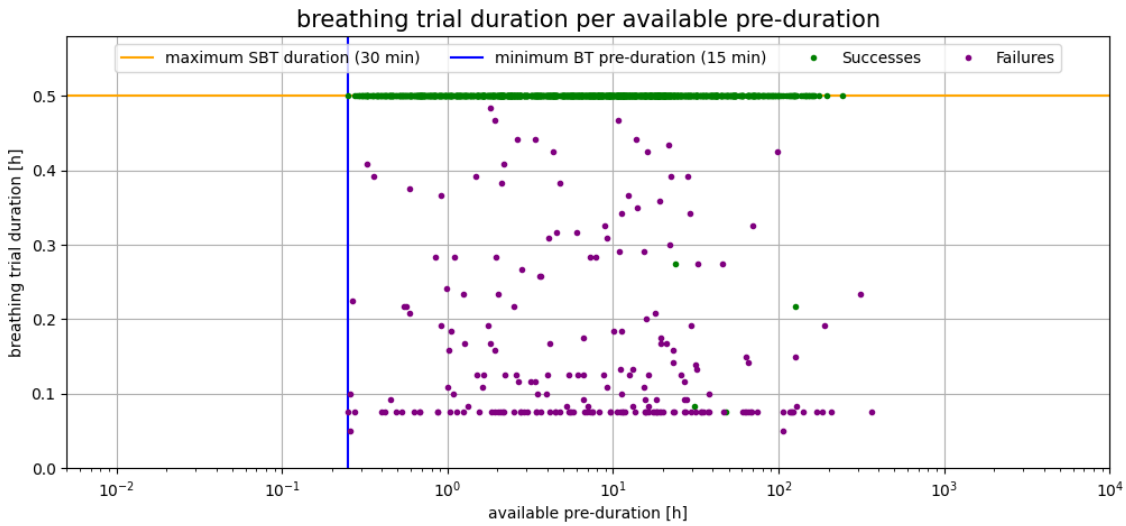

Figure 3.14: Scatter plot of SBT duration vs preduration.

We can also see from Figure (4.2) that the pre-SBT duration can vary drastically. Yet, untraceable SBTs have limited SBT predurations that come from the limited duration of SBT stays. When untraceable SBTs are present, the ratio of traceable and untraceable SBTs seems to stay constant over the different values of pre-SBT duration.
Similarly, the ratio between failures and successful SBTs seems to remain constant, which suggests the lack of correlation between pre-SBT duration and outcome.

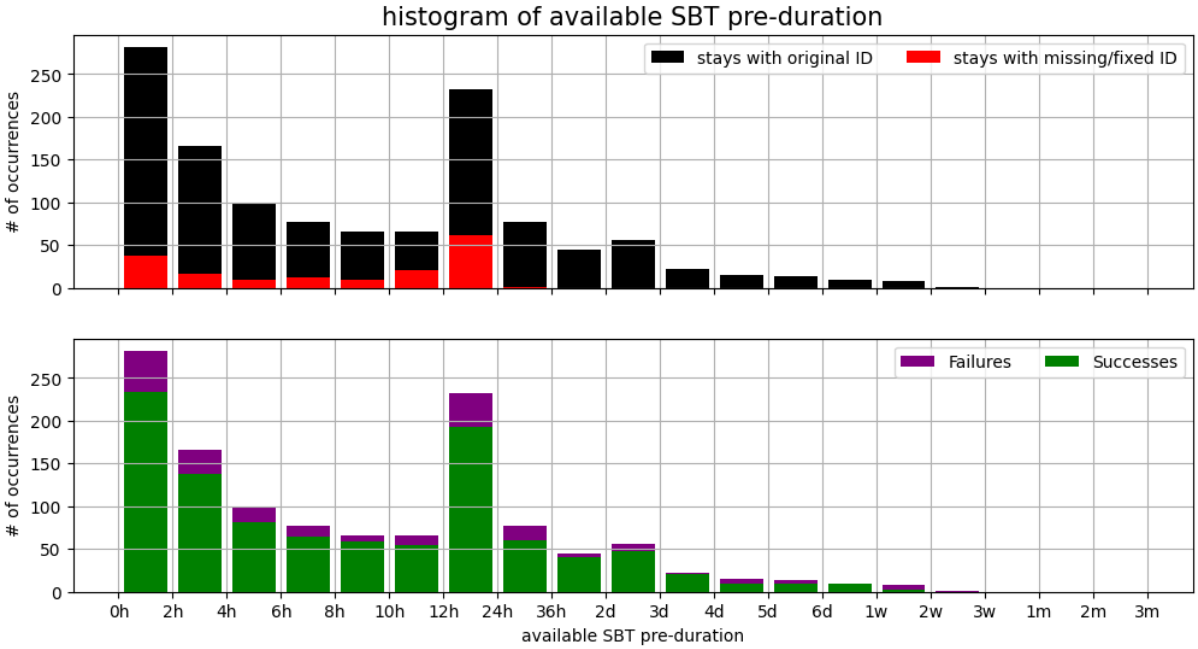

Figure 3.15: Histogram of available pre-SBT duration, highlighting the traceability (on top) and the outcome (on bottom) of the SBTs.

Chapter 4

Machine Learning Architectures

4.1 Feed Forward Neural Network

4.2 Hand-crafted features

We considered 10 physiological signals and only the 15 minutes immediately before each SBT, which corresponded to $p = 300$ data points due to a sampling frequency $f_s = \frac{1}{30s}$.

The simplest feature engineering strategy consisted of extracting the first 3 moments of each medical signal X to characterize the shape of its statistical distribution. To ensure all signals contribute equally to the learning process, the original dataset $\{X\}$ was standardized into $\{Z\}$ (as explained in Section **??**) before extracting the momenta.

- the mean of the data μ represents the center of the distribution

$$\mu = \mathbb{E}[Z] \tag{4.1}$$

- the standard deviation σ measures the amount of variation of the values of a variable about its mean

$$\sigma = \mathbb{E}[(Z - \mu)^2] \tag{4.2}$$

- the skewness γ, specifically the Fisher's moment coefficient of skewness, is a measure of the asymmetry of the probability distribution about its mean

$$\gamma = \mathbb{E}\left[\left(\frac{Z - \mu}{\sigma}\right)^3\right] \tag{4.3}$$

These momenta are meaningful since they are calculated over a single STB, whereas the standardization was applied to the dataset of all SBTs. Given 3 momenta per signal and 10 physiological signals, we obtained 30 features which constitute the inputs of the FFNN.

4.3 PCA-based features

noindent Principal Component Analysis (PCA) is a linear dimensionality reduction technique. The basic idea is to find a new set of orthogonal axes (called principal components) that capture as much of the variability in the dataset as possible. These axes are obtained by identifying directions along which the data vary the most (based on the covariance matrix) and then projecting the data onto those directions.

PCA is variance-sensitive, meaning that features with larger ranges will dominate the analysis and skew the results. For this reason, PCA is typically performed over standardized inputs $Z \in \mathbb{R}^{n \times p}$ obtained from the original input $X \in \mathbb{R}^{n \times p}$ as:

$$z = \frac{x - \mu_x}{\sigma_x} \quad \forall x \in X \quad \text{where} \quad \mu_x = \frac{1}{|X|} \sum_{x \in X} x \quad , \quad \sigma_x = \sqrt{\frac{1}{|X|} \sum_{x \in X} (x - \mu_x)^2} \tag{4.4}$$

where n is the number of data points and p is the number of features. The standardized inputs Z are used to compute the covariance matrix $\Sigma \in \mathbb{R}^{p \times p}$:

$$\Sigma = \frac{1}{n-1} Z^T \cdot Z \tag{4.5}$$

The principal components are the eigenvectors $\{v_i\}_{i=1}^{p}$ of Σ, and the amount of variance explained by each component v_i is given by the corresponding eigenvalue λ_i.

To reduce dimensionality, we retain the first k components, ordered by decreasing eigenvalue, and arrange them into a projection matrix $W_k \in \mathbb{R}^{p \times k}$:

$$W_k = [v_1, \ldots, v_k] \tag{4.6}$$

We can multiply the dataset Z (with p features) by the projection matrix W_k to obtain the reduced dataset $Y \in \mathbb{R}^{n \times k}$ (with k features):

$$Y = Z \cdot W_k \tag{4.7}$$

Geometrically, this operation rotates the coordinate system (so that the axes align with directions of highest variance) and then discards the remaining $p - k$ directions.
Because the discarded directions typically contain mostly noise or redundant information, the projected dataset Y maintains most of the meaningful variation while having a much smaller dimensionality.

At a later stage, PCA was used in a more conventional and unsupervised manner as a dimensionality-reduction technique. After preprocessing and feature engineering, each data point consisted of a large number p values. Working directly with such a high-dimensional input can increase model complexity, slow down training, and amplify noise. To address this issue, we used PCA to compresses the original high-dimensional input into a compact set of orthogonal features that preserve most of the variance while discarding low-variance/noise directions. Specifically, we retained the first k principal components, ordered by decreasing eigenvalue λ_i, and used their projection as the feature representations.

The second feature engineering strategy consisted of Principal Component Analysis (PCA), which was applied as an unsupervised dimensionality-reduction technique as explained in Section **??** (specifically in Subsection **??**).
Similarly to the previous strategy, we considered 10 physiological signals and only the 15 minutes immediately before each SBT, which corresponded to $p = 300$ data points due to a sampling frequency $f_s = \frac{1}{30s}$. The number $k = 30$ of features outputted by PCA was chosen to guarantee compatibility with the FFNN architecture that is used at all stages, which allowed us to compare the effects of the different feature engineering strategies.

4.4 AE/VAE features

CNN based

Autoencoder and Decoder are symmetrical

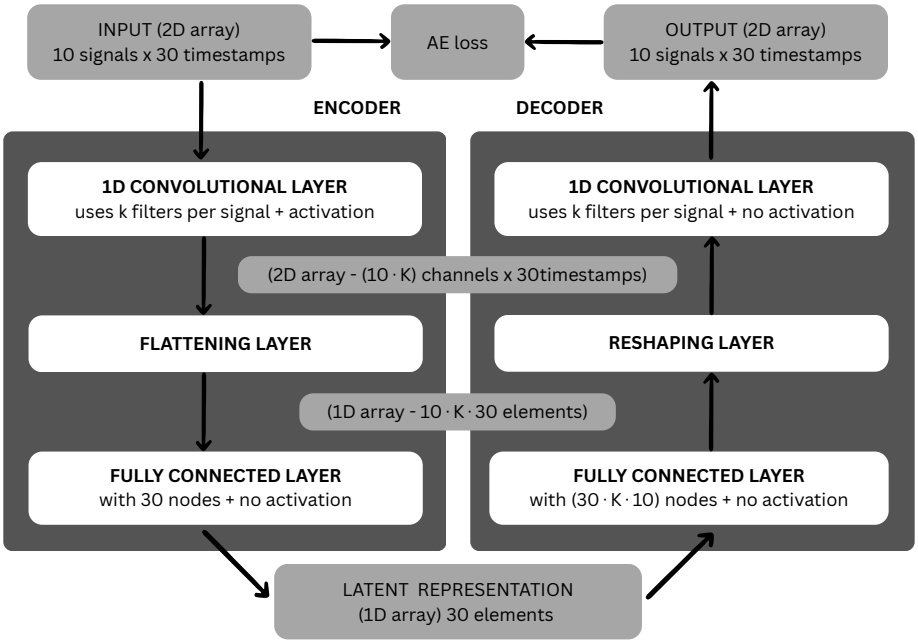

Figure 4.1: Caption.

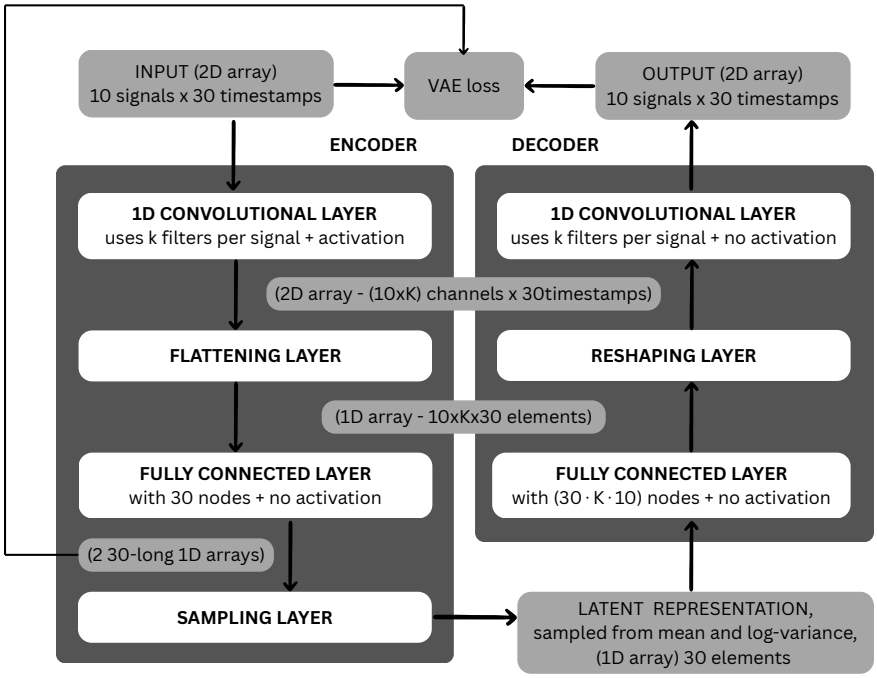

Figure 4.2: Caption.

Chapter 5

Results

5.1 FFNN on hand-crafted features

1

5.2 FFNN on PCA-based features

5.3 FFNN on autoencoder features

5.4 Results Comparison

1

Chapter 6

Conclusions

Appendix

List of all medical signals available

Name	%
Bed	100.0
Timestamp	100.0
Hour	100.0
Date	100.0
HR(Beats/min)	96.27
PVCs/min(Beats/min)	96.06
SpO2 1 Perfusion Index(%)	95.76
SpO2(%)	95.66
SpO2 1 Pulse Rate(Beats/min)	95.61
Ventilator Work Mode	92.26
PPV Source	91.87
Art-M(mmHg)	90.63
Art-S(mmHg)	90.57
Art-D(mmHg)	90.57
ART Pulse Rate - Invasive (Beats/min)	90.56
PPV(%)	89.96
Transthoracic Respiration Rate (Respirations/min)	63.08
VENT_O2_C(%)	62.88
PatientID	61.94
O2%(%)	60.11
Pair(kilopascal (kPa))	59.47
PO2(kilopascal (kPa))	59.47
PEEP_C(cmH2O)	56.52
Trise(s)	56.07
VTe(ml)	54.91
VTi(ml)	54.91
Pmean(cmH2O)	54.91
ftot(Respirations/min)	54.91
PIP(cmH2O)	54.91
fspn(Respirations/min)	54.91
PEEP_M(cmH2O)	54.89
MVspn(l/min)	54.88
MVi(l/min)	54.88
MVe(l/min)	54.88
Plimit(cmH2O)	53.34
F Triger(l/min)	52.99
WOB(J/L)	52.61
Cdyn(ml/cmH20)	52.52
Exp. Flow(l/min)	50.99
P0.1(cmH2O)	48.08
I:E Setting	43.29
I:E	42.93
Rexp(cmH2O/ls)	42.55
Tinsp(s)	42.09
Trise%(%)	41.85

Name	%
VT(ml)	41.35
Pplat(cmH2O)	40.66
TPause%(%)	40.38
Rinsp(cmH2O/ls)	40.36
MV_C(l/min)	39.89
VENT_Delta_Psupp(cmH2O)	38.18
fapnea(Respirations/min)	37.45
Leak Compensation(%)	36.38
Tapnea(s)	35.59
TI_TTOT	34.06
RSBI(l/min)	31.72
T1, Skin(°C)	30.00
T1, Axil(°C)	28.20
f_C(Respirations/min)	27.72
VENT_Flow_C(l/min)	25.89
VT/kg(ml/kg)	19.07
Peep Tot (cmH20)	17.02
NIBP-S(mmHg)	16.75
NIBP-M(mmHg)	16.75
NIBP-D(mmHg)	16.75
NIBP PR(Beat/min)	16.75
T1, Foley(°C)	15.57
Peso (Kg)	15.37
Altezza (cm)	14.86
fCMV (resp/min)	14.1
TPlat(s)	13.83
+Cstat (ml/cmH20)	13.13
pArt-M(mmHg)	9.74
pART Pulse Rate - Peak (Beats/min)	9.70
HR - Hemo (Beats/min)	9.69
pArt-D(mmHg)	9.64
pArt-S(mmHg)	9.64
dPmx(mmHg/s)	9.63
PPV - Hemo (%)	9.60
T Sangue(°C)	9.53
T2, Skin(°C)	7.92
T2, Axil(°C)	7.83
CVP1 Mean	7.07
Pump3 ID	6.69
Pump3_Infusion Status	6.69
Pump3_Total delivered fluid volume(ml)	6.69
Pump3_Infused volume(ml)	6.69
Pump3_Fluid pressure(mmHg)	6.69
Pump3_Fluid delivery time(min)	6.69
Pump3_Bolus Volume(ml)	6.69
SV(ml)	6.64

Name	%
CCO(L/min)	6.64
CPO(W)	6.64
SVV(%)	6.63
CCI(L/min/m²)	6.57
CPI(W/m²)	6.57
SVI(ml/m²)	6.57
VENT_Delta_Pinsp	6.44
SVR - Calc (DS/cm5)	5.74
SVRI - Calc (DS?m²/cm5)	5.69
Pump1_InfusionStatus	5.52
Pump1_Fluid pressure(mmHg)	5.52
Pump1_Infused volume(ml)	5.52
Pump1_Bolus Volume(ml)	5.52
Pump1 ID	5.52
Pump1_Fluid delivery time(min)	5.52
Pump1_Total delivered fluid volume(ml)	5.52
Pump2_InfusionStatus	5.16
Pump2_Fluid delivery time(min)	5.16
Pump2 ID	5.16
Pump2_Bolus Volume(ml)	5.16
Pump2_Fluid pressure(mmHg)	5.16
Pump2_Infused volume(ml)	5.16
Pump2_Total delivered fluid volume(ml)	5.16
Pump4_Fluid delivery time(min)	5.14
Pump4_Total delivered fluid volume(ml)	5.14
Pump4_InfusionStatus	5.14
Pump4_Infused volume(ml)	5.14
Pump4 ID	5.14
Pump4_Bolus Volume(ml)	5.14
Pump4_Fluid pressure(mmHg)	5.14
Pump5_InfusionStatus	5.13
Pump5_Total delivered fluid volume(ml)	5.13
Pump5_Bolus Volume(ml)	5.13
Pump5 ID	5.13
Pump5_Infused volume(ml)	5.13
Pump5_Fluid delivery time(min)	5.13
Pump5_Fluid pressure(mmHg)	5.13
Pump3_Drug Name	4.54
Pump3_Infusion time remaining(min)	4.46
Pump6 ID	4.36
Pump6_Fluid pressure(mmHg)	4.36
Pump6_Bolus Volume(ml)	4.36
Pump6_Infused volume(ml)	4.36
Pump6_Fluid delivery time(min)	4.36
Pump6_Total delivered fluid volume(ml)	4.36
Pump6_InfusionStatus	4.36
+P-Trig (cmH20)	4.29
Pump4_Drug Name	3.86
Pump2_Drug Name	3.83
Pump5_Drug Name	3.70
T2, Foley(°C)	3.57
fSIMV	3.50
Pump1_Drug Name	3.47
Pump6_Drug Name	3.31
Pump4_Infusion time remaining(min)	3.03
PA1 Mean	2.97
PA1 Systolic	2.96
PA1 Diastolic	2.96
Pump2_Infusion time remaining(min)	2.96
Delta Temperature Blood	2.94
Pump1_StandbyTime(s)	2.94
PA1 Pulse Rate	2.94
Pump3_Fluid delivery rate(ml/h)	2.79
Pump1_Infusion time remaining(min)	2.71

Name	%
Pump5_Infusion time remaining(min)	2.70
Pump6_Infusion time remaining(min)	2.30
Pump1_Fluid delivery rate prop(ml/h)	2.26
Pump3_StandbyTime(s)	2.23
Pump5_StandbyTime(s)	2.11
Pump1_Fluid delivery rate(ml/h)	2.03
Pump3_Fluid delivery rate prop(ml/h)	1.97
Pump2_Fluid delivery rate prop(ml/h)	1.91
Pump2_StandbyTime(s)	1.91
Pump4_StandbyTime(s)	1.9
Pump4_Fluid delivery rate prop(ml/h)	1.84
Pump2_Concentration	1.84
Pump2_Dose Rate	1.83
Pump2_Fluid delivery rate(ml/h)	1.60
Pump1_Volume remaining to be infused	1.59
Pump4_Fluid delivery rate(ml/h)	1.59
Pump5_Fluid delivery rate prop(ml/h)	1.56
Pump6_StandbyTime(s)	1.51
fsigh (resp/min)	1.49
Pump6_Fluid delivery rate prop(ml/h)	1.47
QSOFA_TOTAL_SCORE_CLASS	1.43
QSOFA_RR_CLASS	1.43
QSOFA_PSYCHOSIS _CHANGED_CLASS	1.43
QSOFA_BP_S_CLASS	1.43
qSOFA_PSYCHOSIS _CHANGED_STATUS	1.43
Pump5_Fluid delivery rate(ml/h)	1.34
SCORE_QSOFA_BP_S	1.34
Sofa NIBP Systolic	1.34
Pump6_Fluid delivery rate(ml/h)	1.30
Sofa Trigger	1.14
SCORE_QSOFA_RR	1.14
Pump3_Dose Rate	1.13
Pump3_Concentration	1.13
T1(°C)	1.13
EVLW(ml)	1.03
PVPI	1.03
GEDV(ml)	1.03
GEF(%)	1.03
CFI(/min)	1.03
ITBV(ml)	1.03
IBP1 Mean	0.99
IBP1 Dia	0.93
Cardiac Index (Thermodilution)	0.93
IBP1 PR	0.91
IBP1 Sys	0.91
ELWI(ml/kg)	0.87
ITBI(ml/m²)	0.86
GEDI(ml/m²)	0.86
Average CI	0.79
T2(°C)	0.57
Pump5_Concentration	0.44
Pump5_Dose Rate	0.44
Pump4_Dose Rate	0.43
Pump4_Concentration	0.43
Pump6_Concentration	0.31
ICG PAWP	0.3
Pump6_Dose Rate	0.29
+TVapnea (ml)	0.21
Pump1_Dose Rate	0.19
Pump1_Concentration)	0.19
RA Mean (mmHg)	0.16
T1, Core	0.10
IBP 4 Mean(mmHg)	0.06

Table 6.1: List of all medical quantities available in descending order of their abundance (i.e., percentage of files in which the data appears).

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